Probing Warmth and Competence Representations in LLM Hiring Decisions

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Abstract

Recent interpretability work shows that emotions are encoded as linear directions in a language model's residual stream and that steering activations along these directions can alter model behavior. Adapting this emotion-vector method to how a model socially perceives a person, we extract residual-stream directions for warmth and competence, the two axes of the Stereotype Content Model, and examine how these social-intuition vectors relate to callback recommendations across nine open-weight models from the Gemma, Qwen, and Llama families. Warmth and competence are linearly encoded with large effect sizes in every model and shift concept-level judgments under steering, although steering increases callback recommendations in some models and decreases them in others. Internal representations and hiring outcomes show different patterns: both Gemma-3 models track human warmth ratings, whereas the Llama family and earlier Qwen models invert them, and this alignment varies across model generations. In hiring outcomes, models across nearly every family favor names associated with Black and female applicants, contrary to the direction reported in human audit studies; Gemma-3-27B reverses the racial callback gap by more than one standard deviation while remaining the only model to reproduce the human gender direction. We conclude that linear encoding and successful steering do not imply that these representations reliably drive hiring decisions.

Introduction

Work in progress.

Background: Reading a Concept Out of a Language Model

From Layers to Directions. A large language model processes a piece of text by passing it through a long stack of layers, and each layer reads from and writes back to a shared running representation known as the residual stream (Vaswani et al., 2017; Elhage et al., 2021). Think of the residual stream as a notebook passed from one layer to the next: every layer reads what earlier layers have written, adds its own notes, and hands the notebook onward. At any chosen depth, this running state is simply a long list of numbers, a high-dimensional vector, and circuit-level analyses have shown that specific features of the input, from low-level syntax to high-level semantic content, leave consistent geometric traces in that vector across many different contexts (Olah et al., 2020). The idea that such traces take the form of *directions* rather than isolated coordinates goes back to word-embedding analogies, where arithmetic on word vectors captures semantic relationships (Mikolov et al., 2013), and the *linear representation hypothesis* extends this claim to modern transformer models: a high-level feature corresponds to a direction in the residual stream, and projecting the model's state onto that direction yields a scalar measuring how strongly the feature is present (Park et al., 2024). Extracting a direction for some target concept is then a matter of contrast: we collect residual-stream states from texts that express the concept (condition A) and from matched texts that do not (condition B), and take the mean difference between the two groups,

$$v = \bar{h}_A - \bar{h}_B. \tag{1}$$

Normalizing v to a unit vector $\hat{v}$ gives a concept direction, and the concept score of any new piece of text is the projection $\hat{v}^\top h$. The claim worth testing, and the one we validate here through cross-validated classification and topic-holdout accuracy, is that a direction built this way generalizes beyond the handful of texts used to construct it.

Emotion Vectors. A direction built purely from the mean-difference of Eq. 1 only tells us that high- and

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low-condition texts sit in different places; it does not yet tell us whether the model actually relies on that direction when it acts. Testing that requires an intervention rather than an observation: at a chosen layer, we can override the model's own internal state during inference,

$$h' = h + \alpha \, \hat{v}, \tag{2}$$

where α is a signed scalar that sets how strongly, and in which direction, we push. If the model's downstream behavior then shifts systematically as α changes, while an unrelated, orthogonal direction produces no comparable shift, the direction is not merely correlated with the concept but participates causally in the model's computation. Activation addition (Turner et al., 2023) and representation engineering (Zou et al., 2023) established this intervene-and-observe recipe and demonstrated it across a range of models and behaviors. Most recently, Sofroniew et al. (2026) applied exactly this recipe to emotion concepts in Claude Sonnet: building contrastive text pairs for each emotion, they extracted mean-difference directions in the residual stream, pushed the model's state along those directions during inference, and observed systematic, interpretable shifts in the model's behavior. Their study also introduced a neutral-corpus PCA denoising procedure that strips out general evaluative valence from a direction vector, a step we adopt to separate warmth and competence content from shared narrative polarity (detailed in the Supplementary Materials). Fig. 1 illustrates the resulting picture: each emotion occupies its own direction in residual-stream space, with a numeric representation that can be extracted, inspected, and used to steer the model. We reuse this geometry and this method, but not the construct: warmth and competence are social-perception dimensions from the Stereotype Content Model, not emotion states, and calling our direction vectors "emotion vectors" would be a category error.

From Emotions to Hiring. Nothing in this read-and-steer recipe is specific to emotion. Any construct that is linearly encoded in the residual stream should, in principle, be extractable by the same mean-difference procedure and testable by the same steering intervention, whether that construct is an emotional state or a dimension of social perception. Hiring is a domain where such a construct would matter immediately: behavioral audits have already documented that large language models produce uneven callback recommendations across demographic groups (An et al., 2024; Gaebler et al., 2024), yet none of that work locates the representation inside the model that might be doing the work. We therefore ask whether a social-perception construct known to predict human callback outcomes is encoded the same way emotion concepts are, and whether steering it changes a model's own callback recommendations.

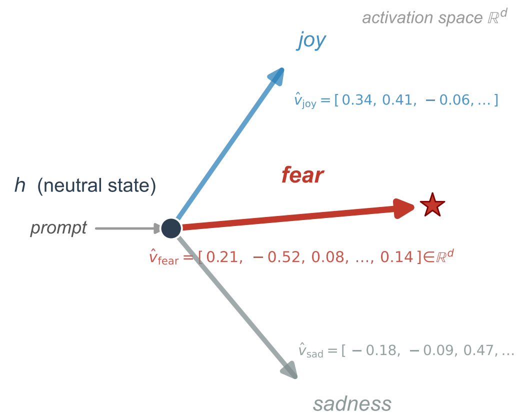

Figure 1: Emotion vectors as directions in activation space (schematic). Conceptually, an emotion vector starts from a neutral model state h. A prompt that evokes a given emotion, such as fear, pushes the residual stream toward a direction $\hat{v}_{\text{fear}}$, encoded here with its representative d-dimensional coordinates. This direction behaves like a commuter who takes the same route to work every day: the attitude, direction, and flow stay the same each time. Across thousands of different prompts that express the same emotion in different words, the model converges on this same direction through its layers, and it produces its output by moving along that direction. After Sofroniew et al. (2026), who call this an emotion vector; we reuse the geometry, not the construct.

The next section describes how we chose that construct and built the materials needed to test it.

Adapting the Method to Hiring

Why Warmth and Competence. The Stereotype Content Model proposes that social perception organizes along two fundamental axes: warmth, capturing perceived intentions and trustworthiness, and competence, capturing perceived ability and effectiveness (Fiske et al., 2002). Evidence across cultural contexts suggests that warmth and competence are recurring dimensions of both interpersonal impressions and perceptions of social groups (Fiske et al., 2007; Cuddy et al., 2008). Warmth and competence therefore give us two well-established, independently motivated axes to search for inside a language model, rather than an ad hoc pair invented for this study.

The Link to Hiring. Gallo et al. (2024) provide the empirical bridge that connects these two axes to real labor-market outcomes. Their released ratings data contain 24,220 warmth and competence judgments from 787 raters covering 282 applicant first names drawn from ten source studies. For the eight name-based correspondence studies with usable callback data, a more favorable first principal component combining warmth and competence was moderately associated with higher callback rates on average. The effect nevertheless varied substantially across studies, and its prediction interval included negative relationships. These findings provide a direct, quantitative connection between the social-perception dimensions we probe and measured labor-market outcomes, while showing that the relationship is not uniform. Behavioral audits of LLMs have separately documented name-driven callback disparities (An et al., 2024; An et al., 2025; Gaebler et al., 2024). To our knowledge, these studies have not tested whether internal warmth and competence representations carry that signal. We use the dataset released by Gallo et al. in two ways: as an external validity check on our probes, and as the reference point against which model-level disparities are compared.

Concept Stories Without Identity. Probing warmth and competence directly from hiring prompts would confound the social-perception dimensions with the hiring decision itself, so the direction vectors have to come from a separate set of materials. We generated 200 short concept stories, evenly split across four conditions (high warmth, low warmth, high competence, low competence), each built around an everyday situation rather than a workplace scenario. Every protagonist in these stories is nameless and gender-neutral, referred to only as "they." This design was not a default choice: pilot work showed that LLM-generated concept stories without this constraint systematically assigned male, Anglo-named protagonists to the low-warmth and low-competence conditions, which would have baked demographic expectations into the direction vectors before any hiring evaluation even began. Removing names and gender from the stories forces each story's activation to be driven by depicted behavior alone, keeping the concept vectors free of the demographic signal we later test for.

Applicant Names. The hiring evaluation itself draws on a separate stimulus set: the 282 applicant first names from Gallo et al. (2024), each inserted into an otherwise identical, fixed job application so that the name is the only signal that varies across trials. Of these 282 names, 149 also appear in the published per-name callback dataset from the same source and carry US race and gender labels; this subset forms the benchmark comparison for demographic disparity. The full set of 282 is used for the probe-versus-human alignment check, since every name carries crowdsourced warmth and competence ratings even when published callback data are unavailable.

Methods

Story Preparation. We generated 200 short narratives (about 90–150 words each), split evenly across four conditions: high warmth, low warmth, high competence, and low competence (50 stories per condition). Stories were written around 50 everyday topics spanning ten domains (workplace, learning, community, sport, travel, and others), with each topic appearing once per condition so that situational vocabulary stays balanced between the high and low poles. Every story uses a nameless, gender-neutral protagonist ("they"), for the reasons given above. Stories were generated by a large language model and checked automatically for forbidden-vocabulary violations, length balance across conditions, and demographic neutrality. This design raises a circularity concern, since the same family of models that wrote the stories is later probed for the concepts those stories express; we return to this point in the Limitations.

Building the Warmth and Competence Vectors. We used TransformerLens (Nanda and Bloom, 2022) to extract residual-stream activations from nine open-weights instruction-tuned models spanning three model families and two version generations each for Gemma and Qwen: Gemma-3-12B-it and Gemma-3-27B-it (Gemma Team, 2025), Gemma-4-12B, Gemma-4-26B-A4B, and Gemma-4-31B; Qwen3-14B (Yang et al., 2025), Qwen3.6-27B, and Qwen3.6-35B-A3B; and Llama-3.1-8B-Instruct (Grattafiori et al., 2024). For each model we captured the residual-stream state at the layer corresponding to about 65 percent of model depth, chosen from a full layer sweep of Cohen's d values (Tab. 1 reports the exact probe layer, model width, and mean residual-stream norm for each model). For each story, activations were mean-pooled across token positions from position 50 onward, skipping the prompt prefix, giving one d_{model}-dimensional vector per story. High- and low-condition stories cluster in distinct regions of this space (Fig. 2), and the warmth and competence direction vectors are simply the mean difference between the two clusters, $v_W = \bar{x}_{\text{high warmth}} - \bar{x}_{\text{low warmth}}$ and $v_C = \bar{x}_{\text{high comp.}} - \bar{x}_{\text{low comp.}}$. All nine models were run under identical conditions (same stories, same pooling rule, same seed 20260527) so that results can be compared across architectures. Every step described in this section, vector building, probe validation, concept- and hiring-level steering, and the hiring evaluation and disparity analysis below, was applied identically to all nine models; complete per-model results for the full suite appear in the Supplementary appendix tables, and Tab. 1 reports parameters for all nine.

We check that each direction is a genuine, generalizable signal rather than an artifact of the specific stories used to build it, using five complementary checks: (i) 5-

Table 1: Model parameters and probe extraction settings for all nine checkpoints. Probe layer is given as selected/total layers. $\|\bar{h}\|$ is the mean residual-stream norm at the probe layer, which sets the scale of steering interventions; the difference across models spans several orders of magnitude, so raw-strength comparisons are not meaningful without normalization.

Model	d_{model}	Probe layer	$\|\bar{h}\|$
Gemma-3-12B-it	3840	31/48	79,722
Gemma-3-27B-it	5376	40/62	61,576
Gemma-4-12B	3840	31/48	97.8
Gemma-4-26B-A4B	2816	19/30	68.4
Gemma-4-31B	5376	39/60	188.5
Qwen3-14B	5120	26/40	206.6
Qwen3.6-27B	5120	42/64	71.9
Qwen3.6-35B-A3B	2048	26/40	3.6
Llama-3.1-8B-Instruct	4096	20/32	11.4

fold cross-validated accuracy, where a logistic regression classifier is trained on full residual-stream representations and scored on held-out folds; (ii) topic-holdout accuracy, a stricter variant using GroupKFold in which all stories from a given topic sit in the same fold, so the probe is tested on entirely unseen situations rather than unseen phrasings of familiar ones; (iii) Cohen's d, the standardized mean difference between high- and low-condition projections onto the direction, compared against an empirical null of 1,000 random unit vectors; (iv) split-half cosine stability, where each condition's 50 stories are randomly split into halves of 25 and the cosine between the two resulting directions is reported; and (v) cross-axis classification, which tests whether the warmth direction predicts competence labels and vice versa, to quantify shared evaluative content between the two axes. We also compute per-story Spearman rank correlations across model pairs to test whether the models converge on a shared cross-architecture construct rather than independent, unrelated ones. A neutral-corpus PCA denoising procedure separates this shared evaluative content from the target concepts; the Supplementary Materials (PCA Denoising) describe the full procedure as a robustness check.

Steering the Concept Vectors. A direction that merely separates high- and low-condition activations does not yet show that the model relies on it; to test that, we apply activation steering (Turner et al., 2023) via TransformerLens hooks at the probe layer (Fig. 2). Steering strength is expressed relative to the mean residual-stream norm at that layer, so that interventions are comparable across models whose activation scales differ by orders of magnitude (see $\|\bar{h}\|$ in Tab. 1). At the concept level, the model answers "Is the protagonist high in warmth/competence? Answer only Yes or No"

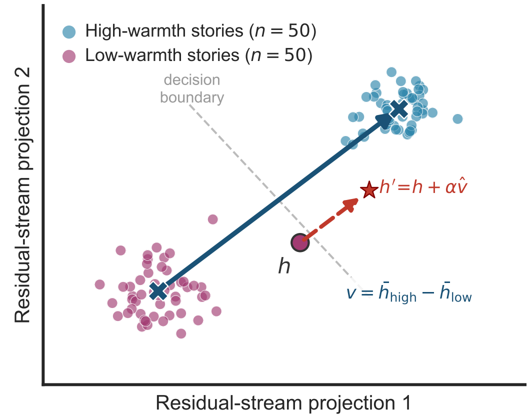

Figure 2: Residual-stream concept geometry (schematic). Low-warmth and high-warmth stories occupy different regions of this space, and the mean-difference direction $v = \bar{h}_{\text{high}} - \bar{h}_{\text{low}}$ (blue arrow) is what separates them. Steering acts like grabbing the wheel of a car that is already driving down one street and turning it until the car crosses onto the next street over: the red dashed arrow, $h' = h + \alpha\,\hat{v}$, shows this same kind of forced redirection, pushing a story's representation from the low-warmth region, across the decision boundary, into the high-warmth region. Concretely, this is done by adding a scaled copy of that same high-warmth direction to the story's current representation. Illustrative synthetic activations, not measured data.

on held-out topics (40 for training, 10 for testing, split by topic index), while we push the residual stream along a local grid of $\alpha \in \{-0.10, -0.05, 0, +0.05, +0.10\}$ times the mean norm. The outcome is the next-token logit margin $\Delta = \text{logit(Yes)} - \text{logit(No)}$. We compare six candidate directions at this stage: the dense target direction, a decoded Gemma Scope 2 SAE direction (Google DeepMind Language Model Interpretability Team, 2025), an axis-specific component, a component shared between the two axes, the opposing concept axis, and a random orthogonal control, since a real effect should be specific to the target direction and absent for an unrelated one. Uncertainty throughout is estimated by paired topic bootstrap resampling.

Steering Hiring Decisions. The hiring prompt pairs each of the 282 applicant first names from Gallo et al. (2024) with an otherwise identical, fixed administrative-assistant application (same education, experience, and skills in every trial), so the name is the only signal that varies. The model answers "would you recommend calling this candidate back for an interview? Answer with a single word: Yes or No," and the outcome is the *callback margin*, the logit difference $\text{logit(Yes)} - \text{logit(No)}$. Race and gender labels are never shown to the model; they are assigned afterward from the Gallo et al. (2024) coding, mirroring the design of human correspondence studies (Bertrand and

Mullainathan, 2004). We then apply the same steering hooks used at the concept level while the model processes these resume-evaluation prompts over a 60-name subset of the applicant roster, using a broad sweep at $\alpha \in \{\pm 0.25, \pm 0.50\}$ and a local-regime follow-up at $\{\pm 0.05, \pm 0.10\}$ for the two Gemma models, so that a push on the warmth or competence direction can be traced directly to a shift in the callback margin.

Names, Human Ratings, and Disparity. Independently of the hiring prompt, each applicant name is embedded in a neutral carrier sentence, and its residual-stream activation is projected onto the warmth and competence directions to yield a per-name probe score. We compare these scores against the aggregated human ratings behind them (24,220 rater judgments from 787 raters across ten source studies) using Spearman rank correlation, which tells us whether the model's internal sense of a name's warmth or competence lines up with how people actually perceive that name. Group-level callback disparities are then computed as differences in mean callback margin between name groups (race: 47 Black-signaling versus 180 White-signaling names; gender: 154 female versus 115 male), standardized by the within-model standard deviation of the callback margin so that models with different logit scales can be compared on equal footing. Human reference gaps come from the published per-name callback rates, aggregated over the 149 names that match the benchmark on first name. Finally, we test whether the internal warmth and competence representations actually carry the name-group signal into the callback decision, using bootstrap mediation along the path name group $\rightarrow$ probe score $\rightarrow$ callback margin (5,000 resamples, seed 20260527, $n = 269$ names with complete data, race subset $n = 227$), reporting indirect effects with 95% confidence intervals for all sixteen model-by-grouping-by-probe combinations.

Results

Discussion and Conclusion

That large language models trained on human-generated text would encode warmth and competence as internally recoverable representations is, in one sense, not surprising: these models distil patterns from the same texts in which warmth and competence shape social perception. What the present results establish empirically is that this encoding is linear, geometrically stable, causally functional at the concept level, and shared across architecturally distinct models from independent laboratories. Each of these properties matters: linearity makes the representation accessible to lightweight probes and steering interventions; causal functionality means the representation participates in explicit judgements about warmth and competence; and cross-architecture convergence suggests the finding reflects a property of training data and the objectives of language modelling rather than a quirk of any single design.

The scale comparison between Gemma-3-12B and Gemma-3-27B complicates this picture in an instructive way. Both models encode warmth and competence with comparable precision, and the 27B model's probe scores correlate slightly more strongly with human warmth and competence ratings. Yet the causal chain from concept representation to hiring decision differs sharply between them. At 12B, increasing the warmth representation produces a large, approximately linear increase in callback inclination (up to +2.37 logit units at $\alpha = +0.10$). At 27B, the equivalent intervention produces a modest positive shift at low strength (+1.97 at $\alpha = +0.05$) that collapses to a large negative shift at slightly higher strength (−2.66 at $\alpha = +0.10$). This non-monotone response is not saturation: the 27B baseline already sits at $P(\text{Yes}) = 0.76$, leaving room for upward movement, and the response is a sign reversal rather than a plateau. A more plausible account is that the 27B decision layer is organised by multiple interacting representations; the concept direction, when amplified beyond a narrow window, begins to interfere with those interactions rather than reinforcing them.

A related puzzle emerges from the cross-model steerability analysis. Among the models with local-regime mediation estimates, Gemma-3-12B has the largest normalized warmth steerability (0.236), yet it shows no significant hiring-level mediation path. Llama-3.1-8B has the smallest value within that subset (0.029) and shows the strongest hiring-level indirect effect (bootstrap IE = +0.190). This steerability paradox illustrates that the capacity to shift a model's representations via linear probing does not imply that those representations causally govern the downstream decision. Models in which representations are most cleanly isolable may be precisely those in which decision processes operate through parallel mechanisms that bypass the steering target.

The warmth anti-alignment observed in Llama-3.1-8B and Qwen3-14B adds a further layer of complexity. Despite high cross-model story agreement (per-story Spearman $\rho \approx 0.76$–0.77 between Gemma and each of the other models), the direction that Llama and Qwen organise as "high warmth" in activation space is negatively correlated with human warmth ratings ($\rho = -0.300$ and -0.193 respectively). The models converge on the same relative ordering of stories yet that shared ordering is inverted relative to the human scale. We interpret this as evidence that instruction tuning has restructured the warmth axis in these architectures: reinforcement learning from human feedback may have

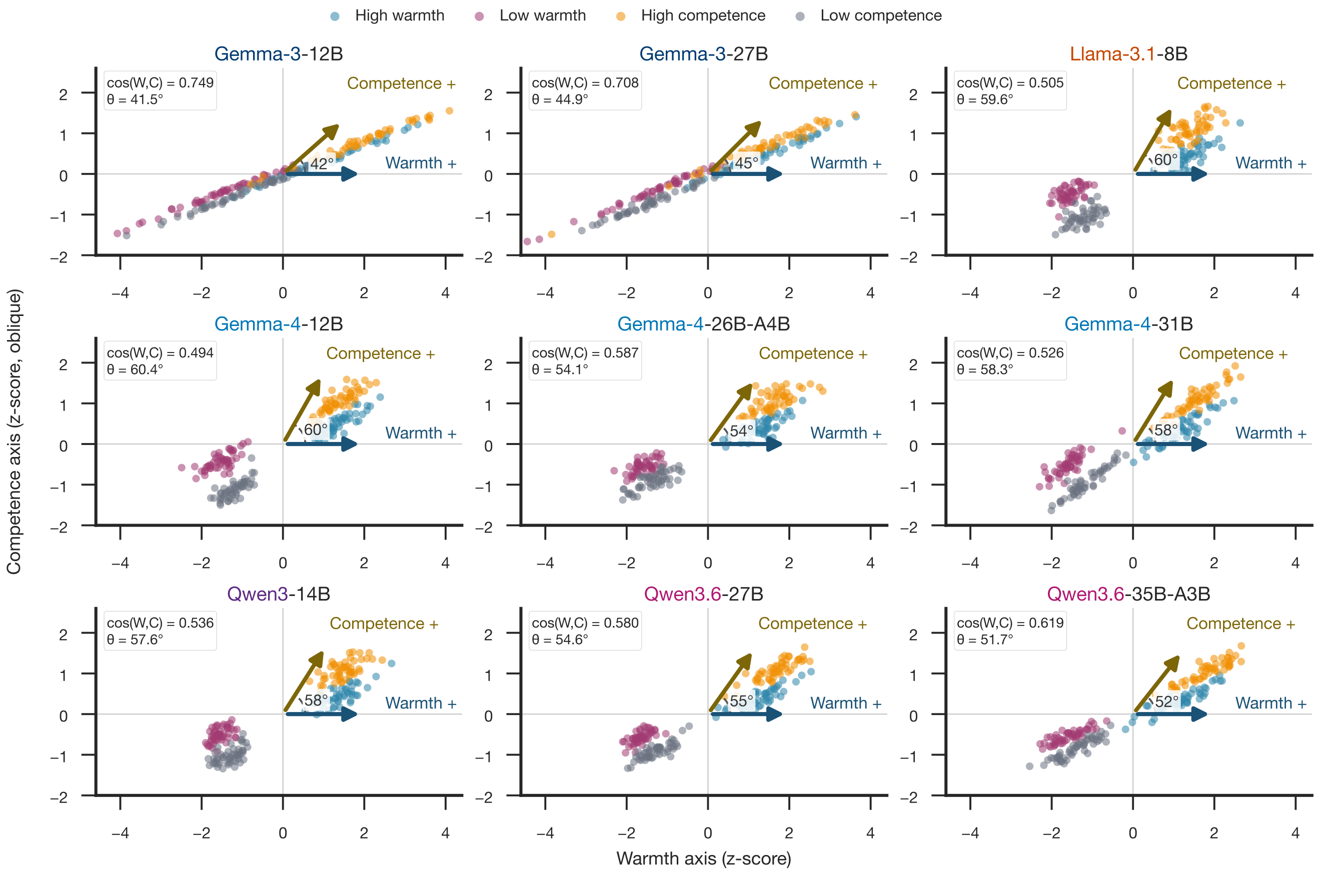

Figure 3: Warmth and competence directions share model-dependent geometry. Each panel projects the same 200 synthetic stories onto raw dense probe-layer directions in an oblique basis. The arrow angle equals $\arccos[\cos(v_W, v_C)]$, so smaller angles indicate stronger geometric alignment. All panels use model-internal z-scores, shared axis limits, and equal x/y scaling.

adjusted representations associated with stereotypically warm social groups in a direction that happens to invert the relationship to the original pretraining warmth axis.

The demographic disparity results at Gemma-3-27B illustrate the practical consequence of this representational reorganisation. The model strongly advantages Black-signalling names relative to White-signalling names (+1.18 SD), directly reversing the classical correspondence-study finding. The gender gap direction, by contrast, reproduces the human pattern. The asymmetry between race and gender suggests that instruction tuning has applied differential corrections to different demographic axes, producing a model that is not more equitable overall but differently inequitable: the racial hierarchy is reversed rather than eliminated, and at substantially larger magnitude than the original human disparity. This finding points to a fundamental limitation of outcome-level bias mitigation: correcting behavioral outputs without understanding the representational substrate can shift rather than remove the underlying imbalance.

Together, the results suggest that LLMs have internalised a structure of social perception broadly analogous to the SCM, emerging from training on human-generated text rather than through any explicit design choice. The connection from that internal structure to actual hiring decisions is not, however, simple or consistent across models. This finding points to a methodological implication for bias auditing: probing internal representations and measuring output disparities are complementary but not interchangeable, and a model that performs well on one measure may not perform well on the other. It also opens a path to addressing the problem at the source by intervening on the representation rather than filtering outputs after the fact, and provides a methodological template for connecting the computational tools of mechanistic interpretability to the empirical tradition of correspondence studies in the social sciences.

Limitations. The concept story corpus was generated by a large language model, raising concerns about circu-

Table 2: Name-level correlation between probed warmth/competence and human ratings. For each of the 282 rated applicant names, the model's residual-stream activation at the probe layer is projected onto the warmth and competence direction vectors. "Probe layer" gives the selected layer index and its resolved fraction of total depth (targeted probe_layer_frac = 0.66; the reported fraction is the integer-rounded layer divided by model depth). Warmth/competence columns correlate the probe projection with human warmth/competence ratings; callback columns correlate the probe projection with the model's own unsteered callback margin. All correlations are Spearman ρ. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$, n.s. not significant.

Model	Probe layer (frac)	ρ(warmth, human)	ρ(competence, human)	ρ(callback, warmth)	ρ(callback, competence)
Gemma-3-12B	31 (0.65)	+0.366***	+0.239***	+0.150*	+0.149*
Gemma-3-27B	40 (0.65)	+0.396***	+0.272***	-0.160**	-0.156**
Llama-3.1-8B	20 (0.62)	-0.300***	-0.062 n.s.	+0.092 n.s.	+0.053 n.s.
Gemma-4-12B	31 (0.65)	+0.020 n.s.	+0.222***	-0.110 n.s.	-0.124*
Gemma-4-26B-A4B	19 (0.63)	-0.204***	-0.044 n.s.	+0.510***	+0.373***
Gemma-4-31B	39 (0.65)	+0.267***	+0.293***	+0.024 n.s.	+0.337***
Qwen3-14B	26 (0.65)	-0.193**	+0.465***	+0.195**	+0.426***
Qwen3.6-27B	42 (0.66)	+0.186**	+0.250***	+0.302***	+0.220***
Qwen3.6-35B-A3B	26 (0.65)	+0.211***	+0.131*	+0.344***	+0.197***

larity in a study that probes language model representations. We note, however, that the generating model (Claude, Anthropic) and the probed models (Gemma, Qwen, Llama) come from different organisations and architectures, so the warmth/competence contrasts are not trivially reproduced by a shared generator. Nevertheless, the possibility that AI-generated stimuli share stylistic properties that inflate probe accuracy across model families cannot be fully excluded; replication with human-authored or template-based stimuli is desirable. The corpus covers 50 topics; broader topic coverage and larger per-condition samples would further test how far the directions generalize. The probe-to-human validation correlations are moderate (warmth $\rho = +0.366$ and $+0.396$ for 12B and 27B; competence $\rho = +0.239$ and $+0.272$), indicating that the model's internal probes capture a component of human social perception rather than replicating it faithfully. The warmth anti-alignment in Llama-3.1-8B ($\rho = -0.300$) and Qwen3-14B ($\rho = -0.193$) means that for these models the mean-difference direction captures a construct that is systematically inverted relative to the human warmth scale; results for these models should be interpreted with this inversion in mind. The probe layer for every model is fixed by a single heuristic, probe_layer_frac = 0.66, rather than selected for maximum warmth/competence separability. A full-depth layer sweep on the training story corpus shows that this heuristic layer rarely coincides with the layer of peak separability: at Gemma-3-12B the selected layer reaches a warmth Cohen's d of 2.68, while the best-separated layer in the same sweep reaches 6.27; Gemma-3-27B (2.95 versus 5.10) and Gemma-4-31B (7.56 versus 11.49) show comparable gaps. This mismatch does not fully explain the weak or inverted name-level correlations reported above. Gemma-4-12B already reaches a high separability score at its selected layer ($d = 8.46$), yet the resulting warmth probe fails to correlate with human name ratings ($\rho = +0.020$, n.s.). Story-level separability, which guides layer selection, therefore does not guarantee that the resulting direction generalises to bare applicant names, a case the training stories never present. Whether averaging or weighting probe projections across the several most separable layers would recover a stronger name-level signal than the single fixed layer is a question we leave open. The causal warmth steering effect at the hiring level is positive and strong at 12B but non-monotone and fragile at 27B, so the causal claim does not generalise across scale without qualification. Callback margins are quantised to a 0.125-unit grid due to bf16 precision in the model's output logits; this quantisation is inherent to bf16 inference and is not resolved by float32 subtraction. We attempted a partial fix by casting the Yes/No logits to float32 before subtracting them, which removes rounding in the subtraction step itself, but the two operands are already bf16-quantised by the time they reach that subtraction, so the 0.125 grid persists regardless. A complete fix would require running inference itself in float32, which roughly doubles GPU memory usage and is likely infeasible for the 27B checkpoint on the available cluster hardware; we did not attempt it. At Gemma-3-12B the callback margin standard deviation is 0.14 with only seven unique values, making group-level disparity estimates at 12B unreliable and the 12B R4 results should not be cited as empirical findings. Llama-3.1-8B shows a comparably narrow spread (SD = 0.12, 12 unique values), so its group-level gaps should be read with the same caution. At Gemma-3-27B the larger variance (SD = 0.41, 18 unique values) makes group gaps of approximately 0.5 SD detectable, but the grid structure introduces discrete steps in all reported gap magnitudes. Qwen3-14B (SD = 0.35, 17 unique values) falls in the same usable range as 27B. The sixteen mediation tests are presented uncorrected for multiple comparisons; only the Llama race–warmth indirect effect survives a Bonferroni-corrected threshold, and the smaller significant paths should be treated as suggestive rather than confirmed. The hiring prompt is simplified relative to real-world screening contexts (single role, fixed qualifications), and results may differ under richer evaluation settings or different job types. Finally, the architectural source of the elevated warmth/competence vector cosine observed in Gemma-3 relative to Qwen3 and Llama-3.1 across all network depths remains an open question.

Future Work. The immediate next step is expanding the hiring stimuli to additional job roles to test whether results generalise beyond administrative assistant. The SAE decomposition performed here on Gemma-3 can be extended to Llama-3.1 using the Llama Scope sparse autoencoders (He et al., 2024), enabling a model-family comparison at the feature level. Validating the concept probes against a human-authored warmth/competence dataset would address the circularity concern associated with AI-generated stimuli. On the social-science side, extending the hiring stimuli beyond names to category-level signals (age, disability, religion) would broaden the scope of the benchmark comparison and test whether the representational account generalises beyond race and gender. Translating these findings into practical debiasing interventions and evaluating their robustness to adversarial prompting is a natural next step for applied work.

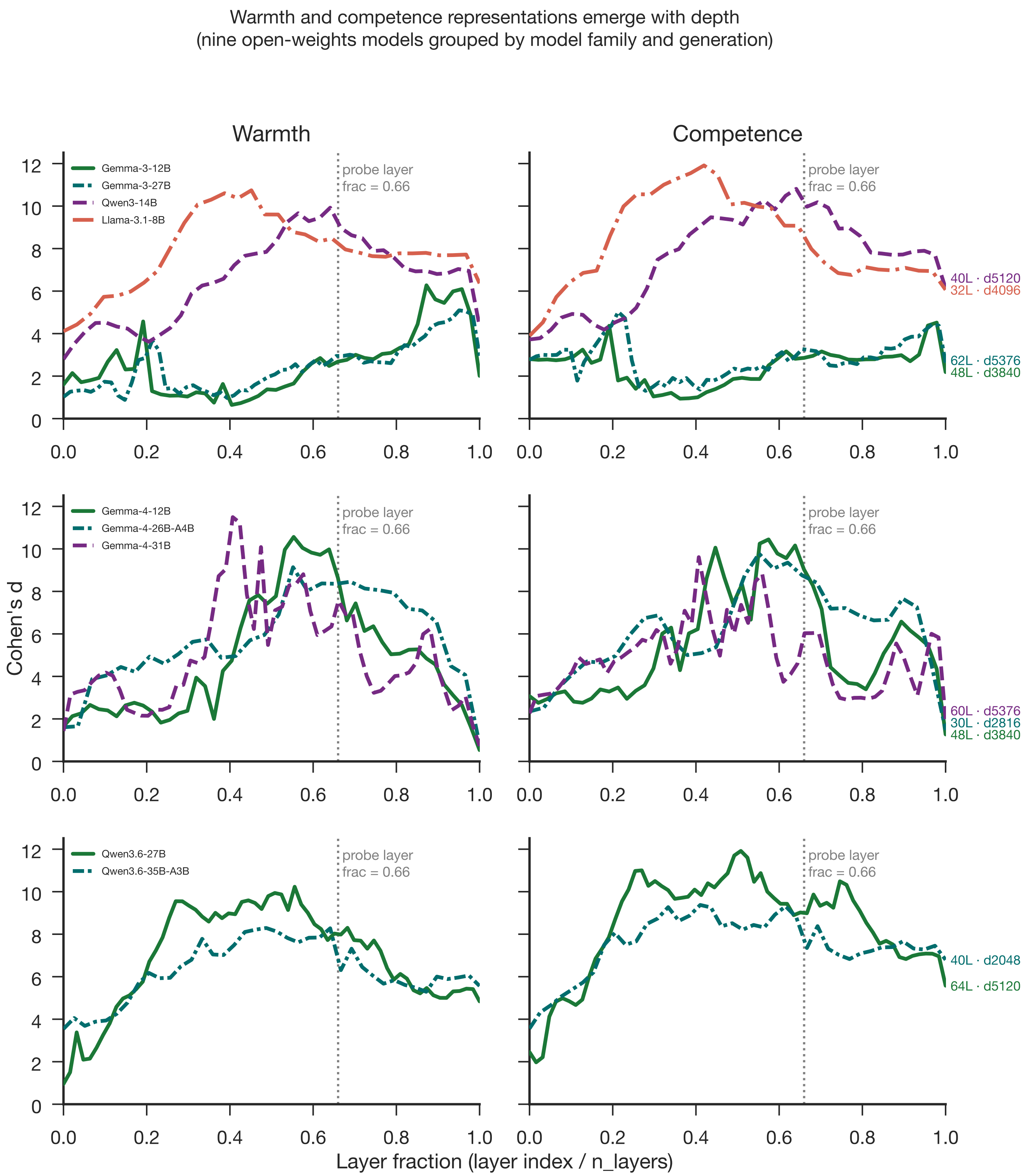

Figure 4: Warmth and competence directions emerge robustly across depth. Layer-wise Cohen's d across nine open-weights models. The dotted vertical line marks the fixed probe-layer fraction of 0.66.

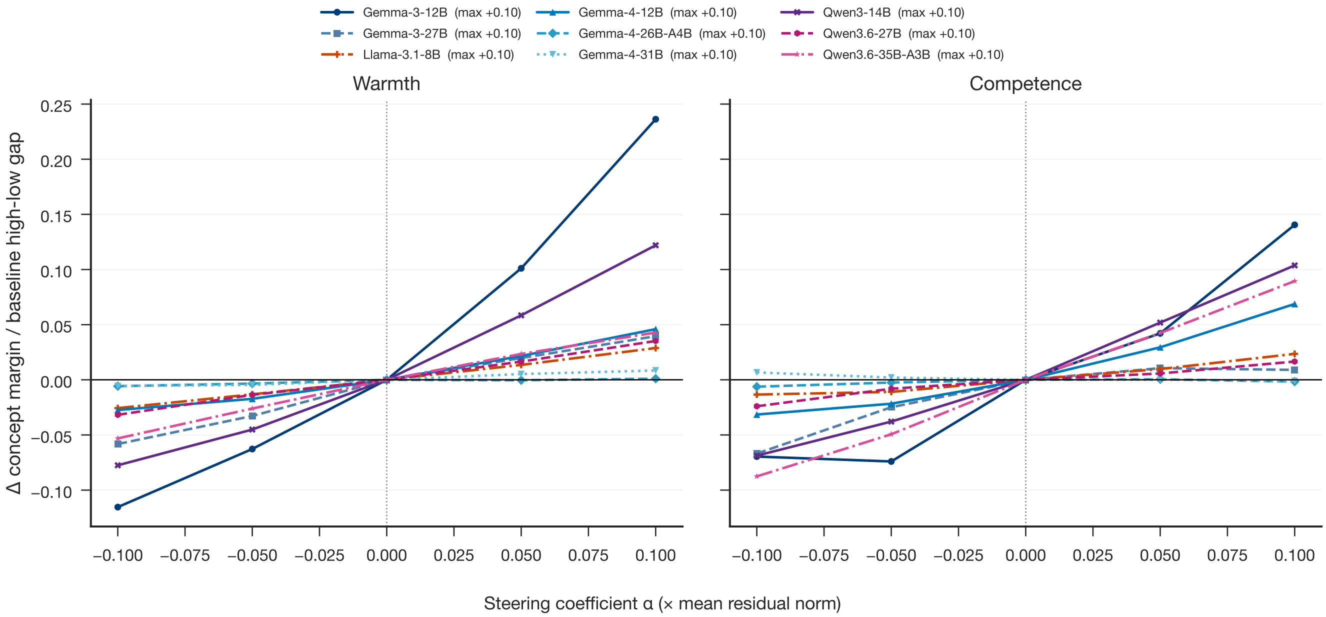

Figure 5: Normalized concept-level steerability varies across nine checkpoints. Each curve shows the additive target-direction change in the held-out Yes-versus-No concept margin divided by the baseline high-versus-low margin gap for the same model and axis. Parentheses in the legend report the maximum positive steering coefficient, $\alpha = +0.10$, expressed relative to the mean residual norm. Normalization reduces raw logit-scale differences but does not establish direction specificity; calibrated random and cross-axis controls are interpreted in the text.

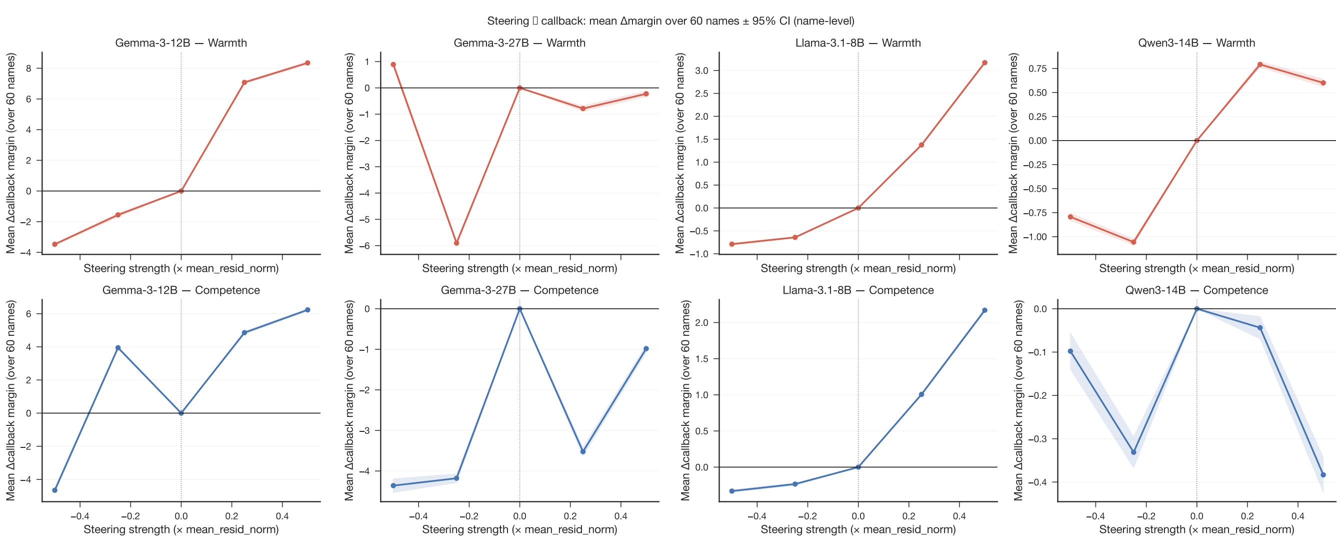

Figure 6: Activation steering shifts hiring callback recommendations. Callback-margin responses to warmth and competence steering in the hiring evaluation setting.

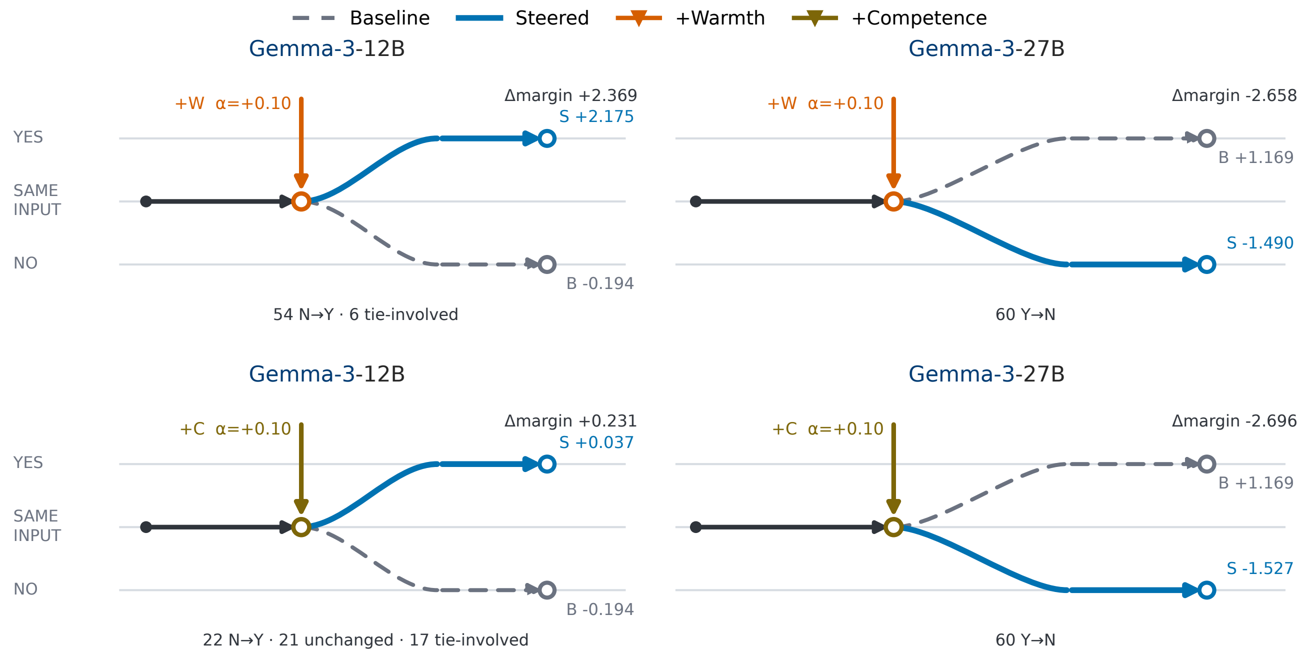

Figure 7: Matched examples show that positive concept steering can move callback decisions in either direction. Rows show warmth and competence; columns compare Gemma-3-12B, whose mean decision moves from No to Yes, with Gemma-3-27B, whose mean decision moves from Yes to No. All panels use the same 60 applications and $\alpha = +0.10$. Gray dashed paths show baseline outcomes, blue paths show steered outcomes, and bottom annotations summarize individual-name transitions. The panels were selected to display both transition directions under matched conditions, not to estimate their prevalence. Endpoint margins, strengths, and transition counts are empirical; connector geometry is schematic. Complete results for all nine models and both axes appear in Tab. S.1.

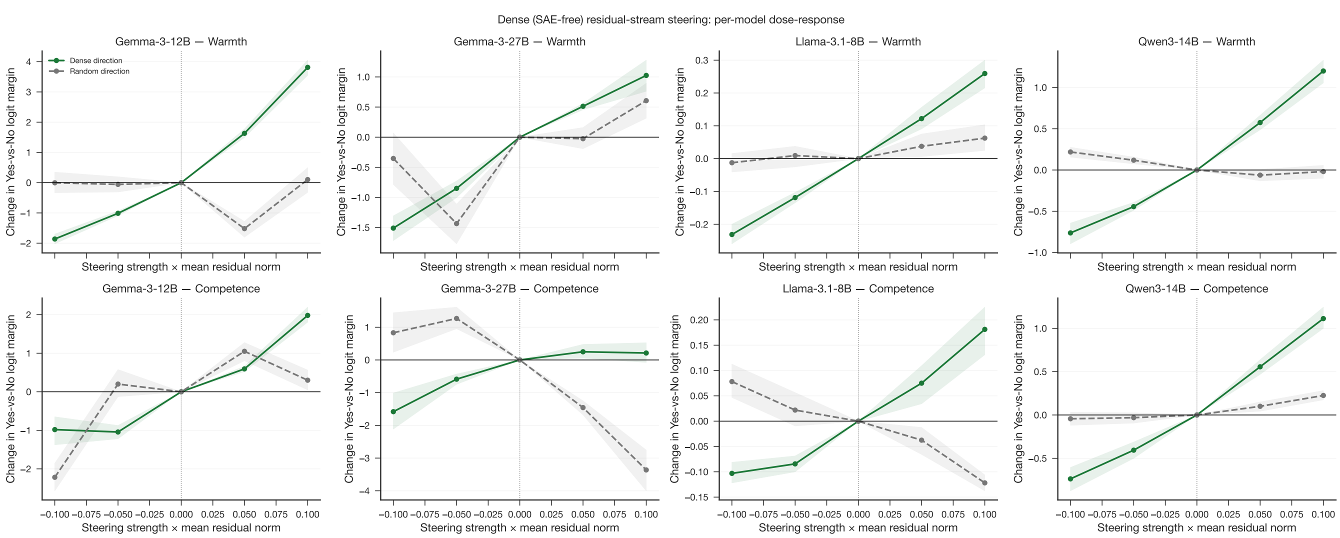

Figure 8: Dense steering dose-response across models. Concept-level dose-response curves for warmth and competence directions.

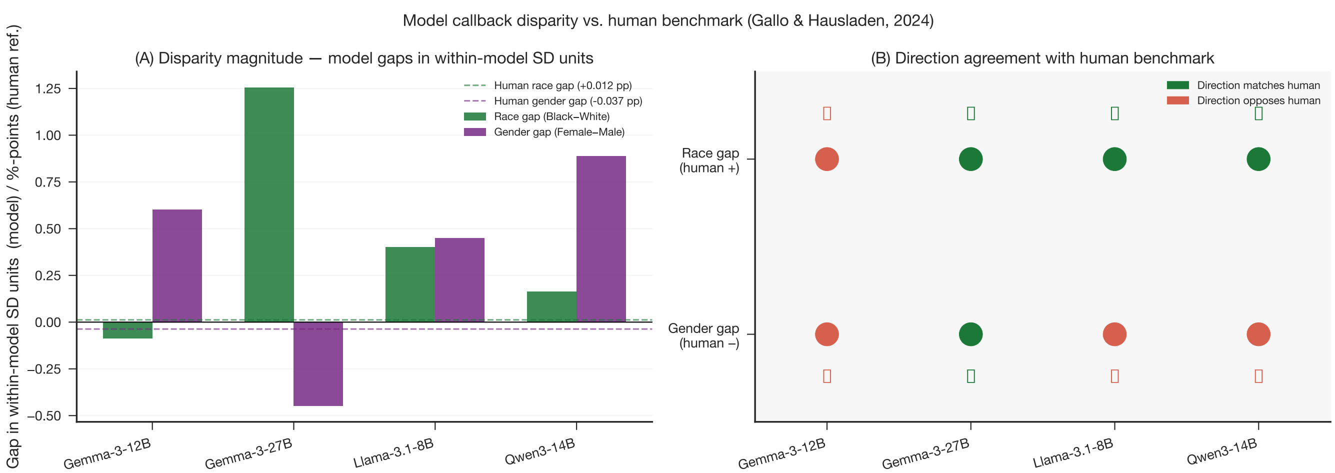

Figure 9: Model callback disparities relative to the human benchmark. Race and gender callback-margin gaps for the model evaluations compared with the correspondence-study benchmark.

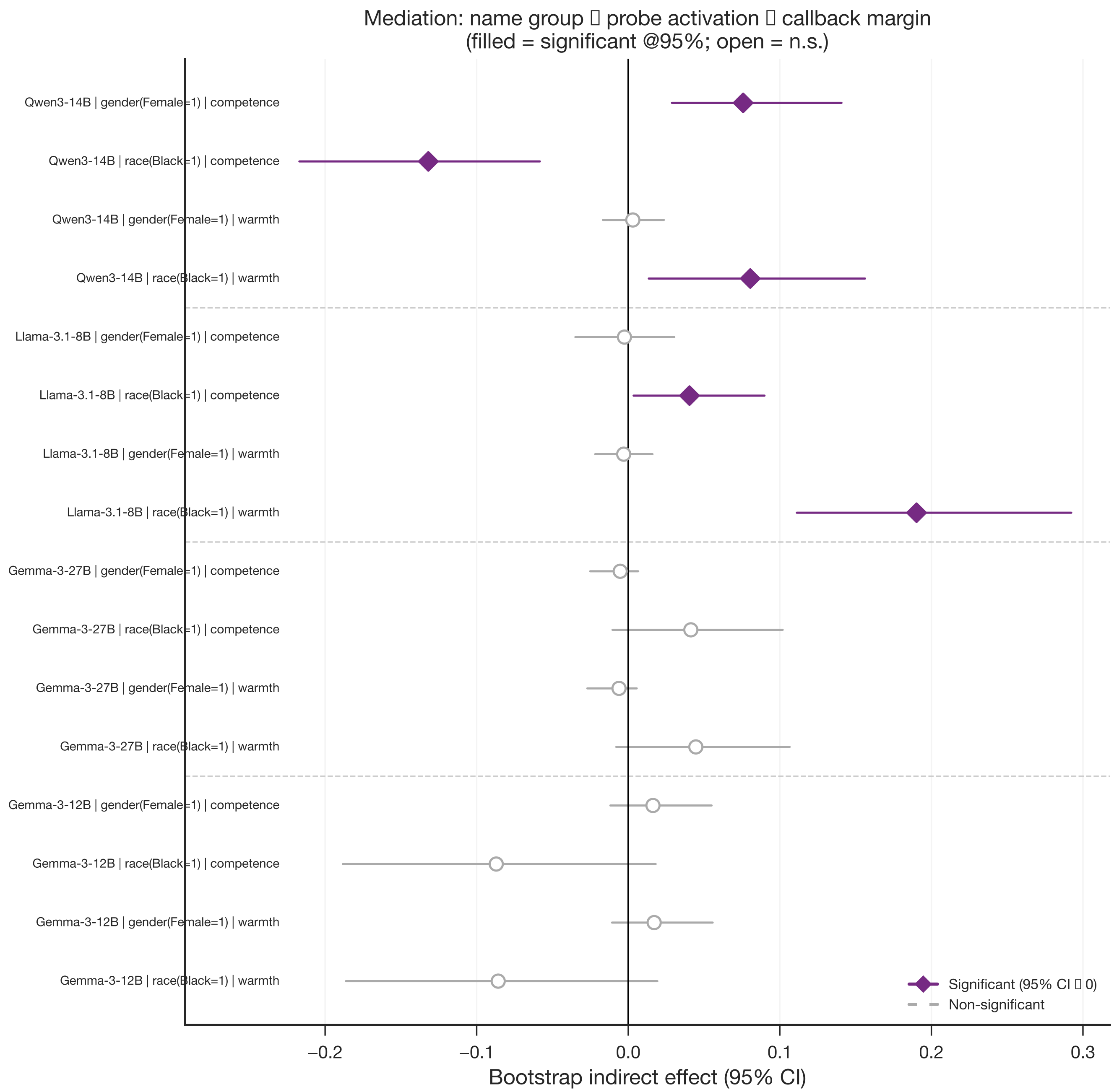

Figure 10: Bootstrap mediation of name-group callback disparities through warmth and competence probe scores. Indirect effects with 95% confidence intervals for the pathway name group → probe score → callback margin; filled markers indicate intervals that exclude zero.

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Supplementary Materials

Details on Methods

Compute and Hardware

All experiments ran on two compute environments. The SCCKN cluster (Universität Konstanz) supplied NVIDIA RTX PRO 6000 Blackwell GPUs (96 GB VRAM), which ran the Gemma-3, Llama-3.1, and Qwen3 jobs (Gemma-3-12B, Gemma-3-27B, Llama-3.1-8B, Qwen3-14B) and most of the extraction and steering jobs for the Gemma-4 and Qwen3.6 checkpoints. Gemma-4-12B is a partial exception: its Stage 1 activations were first extracted on an NVIDIA L40, and a later retry ran on an L40S instead. A dedicated reproducibility audit on the exact L40 hardware class found that both device classes agree on which layer peaks for warmth and competence (layers 26 and 27, respectively), but differ non-negligibly at some layers (maximum competence Cohen's d difference of 0.914, cosine difference of 0.095). The exact-L40 run, hardware-matched to the original Stage 1 extraction, is the one reported throughout the main text. Select calibrated-steering replications for the Gemma-4 and Qwen3.6 checkpoints (all three Gemma-4 models and both Qwen3.6 models) instead ran on NVIDIA H100 GPUs at the Collective Computational Unit (CCU). The authors acknowledge support by the local computing resources through the core facility SCCKN. The authors additionally acknowledge the Collective Computational Unit (CCU) for providing H100 GPU access used in part of this work.

PCA Denoising

The warmth and competence direction vectors share a substantial common component that reflects general narrative valence: stories that portray a protagonist positively (whether warmly or competently) occupy a similar region of the residual stream, and the same holds for negatively valenced stories. To separate conceptual content from this shared evaluative polarity, we follow the neutral-corpus denoising procedure of Sofroniew et al. (2026). We build a neutral corpus of 1,500 Wikipedia introduction paragraphs, length-matched to the concept stories (90–200 words) and filtered to exclude emotionally charged or violent content, and extract residual-stream activations at the same layer and pooling rule used for the concept stories. We then fit principal component analysis (PCA) on these neutral activations and project out, from both direction vectors, the top components that together explain at least 50 percent of neutral variance. For Gemma-3-12B, the first principal component alone explains 56.1 percent of neutral variance, and removing it lowers the cosine between the warmth and competence vectors from 0.749 to 0.530. For Gemma-3-27B, 43 components are needed to cross the 50 percent threshold (50.2 percent explained), lowering the inter-axis cosine from 0.708 to 0.487. The remaining post-denoising cosine, 0.53 and 0.49 respectively, is close to the genuine warmth-competence correlation observed in human ratings ($r \approx 0.61$; Fiske et al. 2002), which suggests that the residual overlap reflects real conceptual co-variation rather than a failure of the denoising step. The denoised vectors serve as a robustness check; the raw dense directions define the primary cross-model causal comparisons and the callback-transition census reported in the main text.

Additional Results

Complete Positive-Steering Callback Transitions

Tab. S.1 reports every model-axis condition used in the positive-endpoint comparison. The table is the complete result; the matched panels in Fig. 7 are illustrative examples selected under a common family, application set, and steering strength. Gemma-3-12B, Gemma-3-27B, and the Qwen3.6 and Gemma-4 checkpoints use $\alpha = +0.10$. Llama-3.1-8B and Qwen3-14B use their available broad-regime endpoint of $\alpha = +0.50$, so raw effect sizes should not be compared directly across all rows.

Positive steering crosses the mean decision boundary in both directions among the Gemma-3 models and from No to Yes for Llama-3.1-8B. The other six models remain in the Yes category for every name, even when the mean margin decreases. This pattern reflects two distinct mechanisms. First, a concept direction constructed from high-minus-low stories is not constrained to align positively with the callback readout. Second, models beginning far from zero can move substantially without changing the categorical recommendation. The Gemma-3-27B response adds a third complication because its local dose-response reverses between +0.05 and +0.10, indicating a narrow nonlinear intervention regime.

Table S.1: Complete positive-endpoint callback transitions. Mean baseline and steered callback margins are followed by their decision category in parentheses. Transition cells list every nonzero individual-name transition among No (N), Tie (T), and Yes (Y); each cell sums to 60.

			Warmth			Competence		
Model	α	Baseline	Endpoint	Δ margin	Transitions	Endpoint	Δ margin	Transitions
Gemma-3-12B	+0.10	-0.194 (N)	+2.175 (Y)	+2.369	54 N→Y; 6 T→Y	+0.037 (Y)	+0.231	19 N→N, 13 N→T, 22 N→Y 1 T→N, 2 T→T, 3 T→Y
Gemma-3-27B	+0.10	+1.169 (Y)	-1.490 (N)	-2.658	60 Y→N	-1.527 (N)	-2.696	60 Y→N
Llama-3.1-8B	+0.50	-2.098 (N)	+1.073 (Y)	+3.171	60 N→Y	+0.072 (Y)	+2.170	15 N→N; 10 N→T; 35 N→Y
Gemma-4-12B	+0.10	+17.188 (Y)	+18.237 (Y)	+1.049	60 Y→Y	+18.715 (Y)	+1.527	60 Y→Y
Gemma-4-26B-A4B	+0.10	+21.245 (Y)	+21.307 (Y)	+0.062	60 Y→Y	+20.865 (Y)	-0.380	60 Y→Y
Gemma-4-31B	+0.10	+25.615 (Y)	+25.173 (Y)	-0.442	60 Y→Y	+24.801 (Y)	-0.814	60 Y→Y
Qwen3-14B	+0.50	+1.704 (Y)	+2.304 (Y)	+0.600	60 Y→Y	+1.321 (Y)	-0.383	60 Y→Y
Qwen3.6-27B	+0.10	+5.346 (Y)	+6.542 (Y)	+1.196	60 Y→Y	+5.879 (Y)	+0.533	60 Y→Y
Qwen3.6-35B-A3B	+0.10	+3.031 (Y)	+3.998 (Y)	+0.967	60 Y→Y	+3.465 (Y)	+0.433	60 Y→Y

Name-Level Warmth/Competence by Marginal Group

Tab. S.2 breaks the same 282-name probe scores down by race (Black/White) and gender (Female/Male) as separate marginal axes, following Gallo et al. (2024)'s own grouping convention. Model warmth and competence values are raw, unnormalized projections onto the concept direction, so they can only be compared within a model, not across the nine rows.

Name-Level Warmth/Competence by Crossed Race × Gender

Tab. S.3 instead crosses race and gender into four joint cells (Black-Female, Black-Male, White-Female, White-Male), recovering the interaction structure that the marginal breakdown above collapses. Group sizes are small and uneven, from 9 names in each Black cell to 84 in White-Female, so the crossed cell means should be read as descriptive rather than as adequately powered group comparisons.

Bootstrap Mediation, All Nine Models

The main text reports bootstrap mediation for four models (Gemma-3-12B, Gemma-3-27B, Llama-3.1-8B, Qwen3-14B) and builds the steerability-paradox argument on those sixteen tests. Tab. S.4 extends the identical procedure, same bootstrap resamples, same seed, same standardization, to the remaining five models (Gemma-4-12B, Gemma-4-26B-A4B, Gemma-4-31B, Qwen3.6-27B, Qwen3.6-35B-A3B), for a combined thirty-six tests across all nine models. Fourteen of the thirty-six indirect effects exclude zero at the uncorrected 95% level; under a Bonferroni threshold across all thirty-six ($\alpha = 0.05/36$), only the Llama-3.1-8B race–warmth path from the main text survives, so every other entry in this table, old or new, remains suggestive rather than confirmed.

Beyond those four models, the remaining five complicate the paradox in one specific way. Both Gemma-3 models showed no significant mediation path on either axis, which the main text reads as evidence that Gemma's cleanly steerable representations bypass the hiring decision rather than driving it. Gemma-4 breaks that pattern: all three Gemma-4 checkpoints show a significant race–competence indirect effect, and Gemma-4-26B-A4B additionally shows a significant gender–warmth path. Qwen3.6 shows a comparable spread of significant competence and warmth paths across its two checkpoints, consistent with the mediation profile already observed in Qwen3-14B. Read together, these results suggest that the absence of hiring-level mediation may be a property of the Gemma-3 generation specifically rather than of architectures with high concept-steerability in general; the four-model steerability paradox in the main text should be understood as scoped to that subset rather than as a claim about steerable models at large.

Table S.2: Name-level warmth/competence and callback margin by marginal demographic group. Race (Black/White) and gender (Female/Male) are treated as separate marginal axes, following Gallo and Hausladen's own grouping convention. Model warmth/competence are raw (unnormalized) projections onto the concept direction and are not comparable in magnitude across models; only within-model group differences are meaningful.

Model	Group	n	Human callback	Model margin	Model warmth / competence
Gemma-3-12B	Black	47	0.183	-0.213	37605.34 / 40319.84
	White	180	0.171	-0.199	38362.85 / 41133.78
	Female	154	0.145	-0.165	38071.05 / 40816.69
	Male	115	0.181	-0.258	37949.40 / 40692.73
Gemma-3-27B	Black	47	0.183	1.665	26143.95 / 25183.93
	White	180	0.171	1.121	26600.90 / 25607.80
	Female	154	0.145	1.118	26448.04 / 25462.46
	Male	115	0.181	1.312	26333.69 / 25349.97
Llama-3.1-8B	Black	47	0.183	-2.020	-1.27 / -0.16
	White	180	0.171	-2.067	-1.34 / -0.17
	Female	154	0.145	-2.054	-1.31 / -0.16
	Male	115	0.181	-2.107	-1.32 / -0.17
Gemma-4-12B	Black	47	0.183	17.194	12.22 / 18.46
	White	180	0.171	17.172	12.29 / 18.96
	Female	154	0.145	17.208	12.23 / 18.70
	Male	115	0.181	17.175	12.30 / 18.74
Gemma-4-26B-A4B	Black	47	0.183	21.509	1.08 / 5.59
	White	180	0.171	21.095	0.90 / 5.47
	Female	154	0.145	21.464	1.00 / 5.51
	Male	115	0.181	20.902	0.92 / 5.54
Gemma-4-31B	Black	47	0.183	25.808	45.31 / 78.15
	White	180	0.171	25.577	45.52 / 78.77
	Female	154	0.145	25.788	45.34 / 78.40
	Male	115	0.181	25.408	45.43 / 78.38
Qwen3-14B	Black	47	0.183	1.758	-8.46 / 17.42
	White	180	0.171	1.700	-9.14 / 17.75
	Female	154	0.145	1.851	-8.64 / 17.76
	Male	115	0.181	1.537	-8.99 / 17.48
Qwen3.6-27B	Black	47	0.183	5.402	5.22 / 8.10
	White	180	0.171	5.328	5.19 / 8.20
	Female	154	0.145	5.384	5.19 / 8.09
	Male	115	0.181	5.284	5.19 / 8.24
Qwen3.6-35B-A3B	Black	47	0.183	2.981	0.17 / 0.39
	White	180	0.171	3.039	0.18 / 0.40
	Female	154	0.145	3.143	0.18 / 0.39
	Male	115	0.181	2.873	0.17 / 0.39

Table S.3: Name-level warmth/competence and callback margin by crossed race × gender group. Applicant names are joined to Gallo et al. (2024)'s published race/gender labels by lowercase first name and matching study (`src/hiring_r4.py`). Model warmth/competence are raw projections and are not comparable in magnitude across models; only within-model comparisons across the four cells are meaningful.

Model	Race × Gender	n	Human callback	Model margin	Model warmth / competence
Gemma-3-12B	Black-Female	9	0.065	-0.181	37660.74 / 40372.96
	Black-Male	9	0.059	-0.278	37842.95 / 40574.00
	White-Female	84	0.129	-0.171	38486.83 / 41263.31
	White-Male	47	0.179	-0.301	38273.64 / 41035.44
Gemma-3-27B	Black-Female	9	0.065	1.514	26107.43 / 25164.89
	Black-Male	9	0.059	1.778	26274.15 / 25309.69
	White-Female	84	0.129	1.033	26642.11 / 25653.64
	White-Male	47	0.179	1.274	26446.95 / 25463.59
Llama-3.1-8B	Black-Female	9	0.065	-1.979	-1.26 / -0.15
	Black-Male	9	0.059	-2.069	-1.27 / -0.18
	White-Female	84	0.129	-2.067	-1.33 / -0.17
	White-Male	47	0.179	-2.104	-1.34 / -0.17
Gemma-4-12B	Black-Female	9	0.065	17.121	12.19 / 18.40
	Black-Male	9	0.059	17.266	12.22 / 18.51
	White-Female	84	0.129	17.173	12.26 / 18.96
	White-Male	47	0.179	17.108	12.36 / 19.00
Gemma-4-26B-A4B	Black-Female	9	0.065	21.840	1.14 / 5.56
	Black-Male	9	0.059	21.160	0.98 / 5.49
	White-Female	84	0.129	21.275	0.92 / 5.44
	White-Male	47	0.179	20.595	0.86 / 5.44
Gemma-4-31B	Black-Female	9	0.065	25.889	45.20 / 78.05
	Black-Male	9	0.059	25.781	45.41 / 78.30
	White-Female	84	0.129	25.817	45.45 / 78.78
	White-Male	47	0.179	25.037	45.58 / 78.76
Qwen3-14B	Black-Female	9	0.065	1.806	-8.39 / 17.61
	Black-Male	9	0.059	1.667	-8.57 / 17.11
	White-Female	84	0.129	1.839	-9.07 / 17.72
	White-Male	47	0.179	1.391	-9.43 / 17.45
Qwen3.6-27B	Black-Female	9	0.065	5.431	5.22 / 8.00
	Black-Male	9	0.059	5.361	5.30 / 8.20
	White-Female	84	0.129	5.368	5.17 / 8.10
	White-Male	47	0.179	5.181	5.20 / 8.28
Qwen3.6-35B-A3B	Black-Female	9	0.065	3.167	0.17 / 0.39
	Black-Male	9	0.059	2.903	0.17 / 0.39
	White-Female	84	0.129	3.174	0.18 / 0.40
	White-Male	47	0.179	2.809	0.18 / 0.40

Table S.4: Bootstrap mediation of name-group callback disparities through warmth and competence probe scores, all nine models. Path: name group → probe score → callback margin, standardized coefficients, Baron–Kenny/Preacher–Hayes bootstrap (5,000 resamples, seed 20260527, n = 269, race subset n = 227). a is the coefficient of group on probe score; b is the coefficient of probe score on callback margin, controlling for group; IE = $a \times b$ is the indirect (mediated) effect. † marks a 95% bootstrap CI that excludes zero. 14 of 36 tests are significant at this uncorrected threshold; only the Llama-3.1-8B race–warmth path (main text) survives a Bonferroni correction across all 36 tests (α = 0.05/36), so the remaining paths listed here, including all newer-model paths, should be read as suggestive rather than confirmed.

Model	Grouping	Probe	a	b	IE	95% CI
Gemma-3-12B	Race (Black=1)	Warmth	-0.502	+0.171	-0.086	[-0.186, +0.019]
	Race (Black=1)	Competence	-0.501	+0.174	-0.087	[-0.188, +0.018]
	Gender (Female=1)	Warmth	+0.072	+0.236	+0.017	[-0.011, +0.056]
	Gender (Female=1)	Competence	+0.068	+0.240	+0.016	[-0.012, +0.055]
Gemma-3-27B	Race (Black=1)	Warmth	-0.391	-0.114	+0.045	[-0.008, +0.106]
	Race (Black=1)	Competence	-0.383	-0.108	+0.041	[-0.010, +0.102]
	Gender (Female=1)	Warmth	+0.093	-0.066	-0.006	[-0.027, +0.006]
	Gender (Female=1)	Competence	+0.096	-0.055	-0.005	[-0.025, +0.007]
Llama-3.1-8B	Race (Black=1)	Warmth	+0.519	+0.366	+0.190†	[+0.111, +0.292]
	Race (Black=1)	Competence	+0.218	+0.185	+0.040†	[+0.004, +0.090]
	Gender (Female=1)	Warmth	+0.027	-0.112	-0.003	[-0.022, +0.016]
	Gender (Female=1)	Competence	+0.208	-0.012	-0.003	[-0.035, +0.030]
Gemma-4-12B	Race (Black=1)	Warmth	-0.260	-0.109	+0.028	[-0.003, +0.069]
	Race (Black=1)	Competence	-0.515	+0.291	-0.150†	[-0.295, -0.019]
	Gender (Female=1)	Warmth	-0.327	-0.119	+0.039	[-0.004, +0.079]
	Gender (Female=1)	Competence	-0.032	-0.024	+0.001	[-0.015, +0.011]
Gemma-4-26B-A4B	Race (Black=1)	Warmth	+0.476	+0.173	+0.082	[-0.038, +0.194]
	Race (Black=1)	Competence	+0.387	+0.339	+0.131†	[+0.060, +0.231]
	Gender (Female=1)	Warmth	+0.213	+0.181	+0.038†	[+0.007, +0.078]
	Gender (Female=1)	Competence	-0.073	+0.386	-0.028	[-0.076, +0.023]
Gemma-4-31B	Race (Black=1)	Warmth	-0.372	+0.143	-0.053	[-0.164, +0.005]
	Race (Black=1)	Competence	-0.465	+0.494	-0.230†	[-0.483, -0.049]
	Gender (Female=1)	Warmth	-0.094	+0.088	-0.008	[-0.042, +0.002]
	Gender (Female=1)	Competence	+0.010	+0.186	+0.002	[-0.029, +0.026]
Qwen3-14B	Race (Black=1)	Warmth	+0.488	+0.165	+0.081†	[+0.013, +0.156]
	Race (Black=1)	Competence	-0.255	+0.517	-0.132†	[-0.217, -0.058]
	Gender (Female=1)	Warmth	+0.210	+0.014	+0.003	[-0.017, +0.024]
	Gender (Female=1)	Competence	+0.218	+0.347	+0.076†	[+0.029, +0.141]
Qwen3.6-27B	Race (Black=1)	Warmth	+0.070	+0.411	+0.029	[-0.029, +0.106]
	Race (Black=1)	Competence	-0.169	+0.290	-0.049†	[-0.103, -0.011]
	Gender (Female=1)	Warmth	-0.021	+0.415	-0.009	[-0.059, +0.047]
	Gender (Female=1)	Competence	-0.320	+0.383	-0.123†	[-0.206, -0.061]
Qwen3.6-35B-A3B	Race (Black=1)	Warmth	-0.228	+0.454	-0.103†	[-0.169, -0.046]
	Race (Black=1)	Competence	-0.097	+0.354	-0.034	[-0.080, +0.006]
	Gender (Female=1)	Warmth	+0.249	+0.257	+0.064†	[+0.027, +0.113]
	Gender (Female=1)	Competence	+0.124	+0.234	+0.029†	[+0.001, +0.067]